SOLUCIONES ROBLEMAS PROPUESTOS AL FINAL DE CADA CAPÍTULO

Capítulo 2

1. Mean = 3%; standard deviation = 6.84%.
2. Series #1: Mean = 0, standard deviation = 39, skewness = 0.
   Series #2: Mean = 0, standard deviation = 39, skewness = −0.63.
3. Series #1: Mean = 0, standard deviation = 17, kurtosis = 1.69.
   Series #2: Mean = 0, standard deviation = 17, kurtosis = 1.
4. The mean, μ, is

$$\mu = \int_0^6 x \frac{x}{18} dx = \left[\frac{x^3}{3 \times 18}\right]_0^6 = \frac{6^3}{3 \times 18} - \frac{0^3}{3 \times 18} = \frac{6^2}{3^2} = 4$$

   The variance, σ^2, is then

$$\sigma^2 = \int_0^6 (x-4)^2 \frac{x}{18} dx = \frac{1}{18} \int_0^6 (x^3 - 8x^2 + 16x) dx$$

$$\sigma^2 = \frac{1}{18} \left[\frac{1}{4}x^4 - \frac{8}{3}x^3 + 8x^2\right]_0^6 = \frac{6^2}{18} \left(\frac{1}{4}6^2 - \frac{8}{3}6 + 8\right)$$

$$\sigma^2 = 2(9 - 16 + 8) = 2$$

5. We need to find h, such that

$$\sum_{i=0}^{h-1} \delta^i = \frac{1}{2} \sum_{i=0}^{n-1} \delta^i = \frac{1}{2} \frac{1-\delta^n}{1-\delta} = \frac{1-\delta^h}{1-\delta}$$

   Solving, we find

$$0.5(1-\delta^n) = 1 - \delta^h$$

$$\delta^h = 0.5 + 0.5\delta^n$$

$$h \ln(\delta) = \ln(0.5 + 0.5\delta^n)$$

$$h = \frac{\ln(0.5 + 0.5\delta^n)}{\ln(\delta)}$$

Alternatively, the formula for the half-life can be expressed as

$$h = \frac{\ln(0.5) + \ln(1 + \delta^n)}{\ln(\delta)}$$

6. We start by computing decay factors and values for x^2.

t	0	1	2	3	4	5	6	7
x	11	84	30	73	56	58	52	35
δ	0.6983	0.7351	0.7738	0.8145	0.8574	0.9025	0.9500	1.0000
x^2	121	7,056	900	5,329	3,136	3,364	2,704	1,225

For the mean, using Equation 2.25, we have

$$\hat{\mu}_t = \frac{1-\delta}{1-\delta^n}\sum_{i=0}^{n-1}\delta^i x_{t-i} = 0.15 \times 336.86 = 50.04$$

For the variance, using Equation 2.35, we have

$$\hat{\sigma}_t^2 = A\sum_{i=0}^{n-1}\delta^i x_{t-i}^2 - B\hat{\mu}_t^2 = 0.17 \times 19826.75 - 1.15 \times 50.04^2 = 505.18$$

Finally, we can take the square root of our answer for the variance, to get the standard deviation, 22.48.

7. We start by calculating the following values:

t	0	1	2	3	4	5
x	0.04	0.84	0.28	0.62	0.42	0.46
δ_1	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
δ_2	0.9044	0.9135	0.9227	0.9321	0.9415	0.9510
δ_3	0.3487	0.3874	0.4305	0.4783	0.5314	0.5905

t	6	7	8	9	10
x	0.66	0.69	0.39	0.99	0.37
δ_1	1.0000	1.0000	1.0000	1.0000	1.0000
δ_2	0.9606	0.9703	0.9801	0.9900	1.0000
δ_3	0.6561	0.7290	0.8100	0.9000	1.0000

We then use Equation 2.25 to calculate our estimates of the mean: mean (no decay) = 0.5236; mean (decay = 0.99) = 0.5263; mean (decay = 0.90) = 0.5486.

8. We start by expanding the table from our answer to question 7:

t	0	1	2	3	4	5
x	0.04	0.84	0.28	0.62	0.42	0.46
δ_1	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
δ_2	0.9044	0.9135	0.9227	0.9321	0.9415	0.9510
δ_3	0.3487	0.3874	0.4305	0.4783	0.5314	0.5905
x^2	0.0016	0.7056	0.0784	0.3844	0.1764	0.2116
$(x - \mathrm{E}[x])^2$	0.233904	0.100086	0.059359	0.009286	0.01074	0.00405

t	6	7	8	9	10
x	0.66	0.69	0.39	0.99	0.37
δ_1	1.0000	1.0000	1.0000	1.0000	1.0000
δ_2	0.9606	0.9703	0.9801	0.9900	1.0000
δ_3	0.6561	0.7290	0.8100	0.9000	1.0000
x^2	0.4356	0.4761	0.1521	0.9801	0.1369
$(x - \mathrm{E}[x])^2$	0.018595	0.027677	0.017859	0.217495	0.023604

In the last line, we have used our estimate of the mean (no decay) from the previous problem.

For the first estimator with no decay factor, we can use Equation 2.19 to calculate the variance:

$$\hat{\sigma}_x^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \hat{\mu}_x)^2 = \frac{1}{11-1}0.7227 = 0.0723$$

For the second and third estimators, we use Equation 2.35 and our estimates of the mean from the previous question:

$$\hat{\sigma}_t^2 = A\sum_{i=0}^{n-1}\delta^i x_{t-i}^2 - B\hat{\mu}_t^2 = 0.11 \times 3.58 - 1.10 \times 0.5263^2 = 0.0716$$

$$\hat{\sigma}_t^2 = A\sum_{i=0}^{n-1}\delta^i x_{t-i}^2 - B\hat{\mu}_t^2 = 0.16 \times 2.49 - 1.11 \times 0.5486^2 = 0.0681$$

Taking the square root of the variances, we arrive at our final answers: standard deviation (no decay) = 0.2688; standard deviation (decay = 0.99) = 0.2676; standard deviation (decay = 0.90) = 0.2610.

9. The new estimates are 10.10%, 9.82%, and finally 9.78%. These can be found as follows:

$$\hat{\mu}_t = 0.02x_t + 0.98\hat{\mu}_{t-1}$$

$$\hat{\mu}_1 = 0.02 \times 15\% + 0.98 \times 10\% = 10.10\%$$

$$\hat{\mu}_2 = 0.02 \times (-4\%) + 0.98 \times 10.10\% = 9.82\%$$

$$\hat{\mu}_3 = 0.02 \times 8\% + 0.98 \times 9.82\% = 9.78\%$$

10. The half-lives are:

$$h_{200} = \frac{\ln(0.5 + 0.5 \times 0.95^{200})}{\ln(0.95)} = 13.5127$$

$$h_{1,000} = \frac{\ln(0.5 + 0.5 \times 0.95^{1,000})}{\ln(0.95)} = 13.5134$$

11. The half-life of the EWMA estimator is approximately 11.11 days. A rectangular window with 22 days would have the most similar half-life, 11 days.

$$h_{32} = \frac{\ln(0.5 + 0.5 \times 0.96^{32})}{\ln(0.96)} = 11.11$$

12. $1 - 0.96^{50} = 87\%$.

13. We can use our updating rule,

$$\hat{\sigma}_t^2 = (1 - \delta)r_t^2 + \delta\hat{\sigma}_{t-1}^2$$

to calculate successive estimates of the variance. The estimate of the standard deviation is just the square root of the variance estimator:

t	0	1	2	3	4	5	6
r		−5%	18%	16%	−2%	5%	−10%
$E[\sigma^2]$	0.010000	0.009625	0.010764	0.011506	0.010950	0.010528	0.010501
$E[\sigma]$	10%	9.81%	10.37%	10.73%	10.46%	10.26%	10.25%

The final estimate of the standard deviation is approximately 10.25%.

CAPÍTULO 3

1. We take the approximation for dV as a given, as an equality, and take the expectations of both sides:

$$\mathrm{E}[dV] = \tilde{\Delta}\mathrm{E}[\mathrm{R}] + \frac{1}{2}\tilde{\Gamma}\mathrm{E}[\mathrm{R}^2] + \theta \mathrm{dt}$$

Because R has a mean of zero, $\mathrm{E}[R] = 0$ and the variance is equal to $\mathrm{E}[R^2]$.

$$\sigma^2 = \mathrm{E}[R^2] + E[R]^2 = \mathrm{E}[R^2] + 0^2 = \mathrm{E}[R^2]$$

Substituting back into our previous equation, we have

$$\mathrm{E}[dV] = \tilde{\Delta}0 + \frac{1}{2}\tilde{\Gamma}\sigma^2 + \theta \mathrm{dt} = \frac{1}{2}\tilde{\Gamma}\sigma^2 + \theta \mathrm{dt}$$

2. The most likely day for the next exceedance is tomorrow. This is something of a trick question. It is tempting to guess that the next exceedance is equally likely to occur on any future day. In fact, if the probability of an exceedance on any given day is α, then the probability of an exceedance tomorrow is α, but the probability that the *next* exceedance is the day after tomorrow is $(1-\alpha)\alpha$. This is because in order for the *next* exceedance to be the day after tomorrow, two things have to happen. First, there must be no exceedance tomorrow. Second, there must be an exceedance the day after tomorrow. The probability of these two events are $(1-\alpha)$ and α, respectively, so the probability that both happen is $(1-\alpha)\alpha$. Because $(1-\alpha)$is less than one, $(1-\alpha)\alpha<\alpha$. For example, if we are calculating the 95% VaR, then $\alpha = 5\%$. The probability that the next exceedance is tomorrow is 5%, and the probability that the next exceedance is the day after tomorrow is $5\% \times 95\% = 4.75\%$. The probability continues to decline further out. The probability that the next exceedance happens in n day is, $(1-\alpha)^{(n-1)}\alpha$, which is likewise always less than α.
3. Five percent of 256 is 12.8, so the 95% VaR is between the 12th and 13th worst returns. Because we do not know the distribution between the 12th and 13th worst returns, we use the 12th. The one-day 95% VaR is −16%, or a loss of 16%.
4. In order to calculate the hybrid VaR, we need to calculate weights for the returns. For each return, the corresponding weight is 0.99^t. We can convert this to a percentage weight by dividing by the total weight, $92.37 = (1 - 0.99^{256})/(1 - 0.99)$. Starting with the worst return, we then sum these percentage weights to get the cumulative percentage weight until we reach 5%. This time, 5% occurs between the 7th and 8th worst

returns. The 7th worst return, −20%, is our one-day 95% VaR. Many of the worst returns have occurred recently, so it is not surprising that the hybrid VaR is worse than the historical VaR.

	t	***R***	**wt**	**%wt**	**cum. %**
1	42	−35%	0.6557	0.71%	0.71%
2	83	−29%	0.4342	0.47%	1.18%
3	10	−26%	0.9044	0.98%	2.16%
4	23	−25%	0.7936	0.86%	3.02%
5	3	−24%	0.9703	1.05%	4.07%
6	58	−21%	0.5583	0.60%	4.67%
7	188	−20%	0.1512	0.16%	4.84%
8	103	−19%	0.3552	0.38%	5.22%
9	131	−18%	0.2680	0.29%	
10	12	−16%	0.8864	0.96%	
11	116	−16%	0.3117	0.34%	
12	245	−16%	0.0852	0.09%	
13	150	−15%	0.2215	0.24%	
14	56	−14%	0.5696	0.62%	
15	61	−14%	0.5417	0.59%	
16	31	−13%	0.7323	0.79%	
17	69	−13%	0.4998	0.54%	
18	95	−13%	0.3849	0.42%	
19	161	−13%	0.1983	0.21%	
20	35	−12%	0.7034	0.76%	

5. For a normal distribution, 5% of the weight is less than –1.64 standard deviations from the mean. The 95% VaR can be found as: 0.40% – 1.64 × 2.30% = −3.38%. Because of our quoting convention for VaR, the final answer is VaR = 3.38%.

6. To find the 95% VaR, we need to find v, such that:

$$\int_{-100}^{v} p d\pi = 0.05$$

Solving, we have:

$$\int_{-100}^{v} \frac{1}{200} d\pi = \left[\frac{\pi}{200}\right]_{-100}^{v} = \frac{v+100}{200} = 0.05$$
$$v = -90$$

The VaR is a loss of 90. Alternatively, we could have used geometric arguments to arrive at the same conclusion. In this problem, the PDF describes a rectangle with a base of 200 units and a height of 1/200. As required, the total area under the PDF (base multiplied by height) is equal to one. The leftmost fraction of the rectangle, from –100 to –90, is also a rectangle, with a base of 10 units and the same height, giving an area of 1/20, or 5% of the total area. The edge of this area is our VaR, as previously found by integration.

7. To find the 95% VaR, we need to find v, such that

$$0.05 = \int_{15}^{v} p d\pi$$

By inspection, half the distribution is below 5, so we need only bother with the first half of the function,

$$\begin{aligned} 0.05 &= \int_{-15}^{v} \left(\frac{3}{80} + \frac{1}{400}\pi\right) d\pi \\ &= \left[\frac{3}{80}\pi + \frac{1}{800}\pi^2\right]_{-15}^{v} \\ &= \frac{3}{80}(v+15) + \frac{1}{800}\left(v^2 - 15^2\right) \\ &= \frac{3}{80}(v+15) + \frac{1}{800}\left(v^2 - 225\right) \end{aligned}$$

Rearranging terms,

$$v^2 + 30v + 185 = 0$$

We can solve this using the quadratic equation:

$$v = \frac{-30 \pm \sqrt{900 - 4 \times 185}}{2} = -15 \pm 2\sqrt{10}$$

Because the distribution is not defined for $\pi < -15$, we can ignore the negative, giving us the final answer of

$$v = -15 + 2\sqrt{10} \approx -8.68$$

The one-day 95% VaR for Pyramid Asset Management is approximately 8.68.

8. To answer this question, we need to first calculate the delta of the put option. We can do this by using put-call parity. Assume the price of a call is C, the price of a put is P. For a call and put with the same expiration on a non-dividend paying stock, put-call parity states

$$C - P = S - Xe^{-rT}$$

Taking the derivative of both sides with respect to S, we have

$$\Delta_C - \Delta_P = 1$$

Or, rearranging terms,

$$\Delta_P = \Delta_C - 1$$

Assuming the same underlying, same expiration and same strike, as in the question, if the delta of the call option is 0.50, then the delta of the put must be −0.50.

The 95% VaR corresponds to the bottom 5% of returns. For a normal distribution 5% of the distribution is less than 1.64 standard deviations below the mean. We can get this result from a lookup table, a statistics application, or a spreadsheet. For example, in Excel = NORM.S.INV(0.05) would give us −1.64, the negative sign indicating that the result is below the mean. Given the standard deviation of 2%, this corresponds to a −3.28% return for XYZ.

For the call option, the delta-adjusted exposure is $100 × 0.50 = $50. In other words, for small changes in the underlying price, owning one call option is like owning $50 worth of stock. The one-day 95% VaR of the call option is then a loss of $1.64: $50 × −3.28% = −$1.64.

For the put option, the delta-adjusted exposure is $100 × −0.50 = −$50. In other words, for small changes in the underlying price, owning one put option is equivalent to being short $50 worth of stock. Now, if you are long a put, the worst thing that can happen is that the stock price goes up, not down. To calculate the 95% VaR of the put, we shouldn't use the 5th percentile of the normal distribution, but the 95th percentile. Because the normal distribution is symmetric, if the 5th percentile is −1.64 standard deviations from the mean then the 95th percentile is +1.64. So rather than −3.28%, we should use +3.28%. The one-day 95% VaR of the put option is then a loss of $1.64: −$50 × 3.28% = −$1.64. Notice that the call option and put option, because they have the same absolute delta-adjusted exposure both have the same VaR.

For the portfolio as a whole, the VaR is zero. This might seem strange given that the call and put both have positive VaR in isolation, but remember, VaR is not additive. The reason the portfolio VaR is zero is that the delta-adjusted exposure of the portfolio is zero, −$50 + $50 = $0. For small changes in the underlying price, the portfolio value will not change significantly.

In reality, this portfolio is not free of risk. This combination of a put and a call is often referred to as a straddle. It can be thought of as a bet on volatility. The straddle will make money if the underlying price increases or decreases significantly. Conversely, you will lose money if the stock stays near the strike price of the options. Even if the stock price doesn't change, a change in implied volatility could significantly alter the value of the straddle. A straddle can lose money both in the short run and long run. Even though the delta-normal VaR is zero, this portfolio is not risk free.

9. The VaR of the call and the put in isolation are \$1.64, just as in the previous equation. This is because the absolute delta-adjusted exposures are the same. Similarly, the VaR of the portfolio is the same, \$0, because the delta-adjusted exposure is zero.

Just as before, even though the delta-adjusted VaR of this portfolio is zero, this portfolio is not risk free. The portfolios are not equally risky. This portfolio, where we are short both options is much riskier. As a risk manager, alarm bells should always go off when you see a short option position. If you are long a call or put, the worst-case scenario is that the value of the option goes to zero. If you are long an option, the most you can lose is the current value of the option. If you are short a call option, though, the potential loss is unlimited (if you are short a put option, the worst-case scenario is for the underlying price to go to zero; the potential loss is limited, but it could still be very significant). Because of this, the worst-case scenario for a short straddle is worse than the worst-case scenario of a long straddle.Most risk managers would therefore view the short straddle as significantly riskier.

CAPÍTULO 4

1. Positive homogeneity requires that multiplying all outcomes by a constant, c, also increases the risk measure by c. For a random variable, X, the variance is defined as

$$\sigma_X^2 = \mathrm{E}[(X - E[X])^2]$$

Multiplying all the outcomes of X by c is the same as multiplying X by c. If we multiply X by c then we have

$$\begin{aligned}
\sigma_{cX}^2 &= E[(cX - E[cX])^2] \\
&= E[(cX - cE[X])^2] \\
&= \mathrm{E}[(c(X - E[X]))^2] \\
&= E[c^2(X - E[X])^2] \\
&= c^2 E[(X - E[X])^2] \\
&= c^2 \sigma_X^2
\end{aligned}$$

Taking the square root of both sides, we have

$$\sigma_{cX} = c\sigma_X$$

The standard deviation of cX is c times the standard deviation of X. This proves that standard deviation displays positive homogeneity.

2. The minimum expected shortfall of the portfolio is −$700. Because expected shortfall is subadditive, we know that the expected shortfall of the portfolio will be at worst the sum of the expected shortfalls of the two securities separately.

3. In the previous chapter we found that the VaR for Box Asset Management was equal to a loss of 90. To find the expected shortfall, we can use Equation 4.3 as follows:

$$\begin{aligned} S &= -\frac{1}{1-\gamma}\int_{-\infty}^{VaR} xf(x)dx \\ &= -\frac{1}{0.05}\int_{-100}^{-90} \pi\frac{1}{200}d\pi \\ &= -\frac{1}{20}[\pi^2]_{-100}^{-90} \\ &= 95 \end{aligned}$$

The final answer, a loss of 95 for the expected shortfall, makes sense. The PDF in this problem is a uniform distribution, with a minimum at –100 or a loss of +100. Because it is a uniform distribution, all losses between the VaR, 90, and the maximum loss, 100, are equally likely; therefore, the mean loss, given a VaR exceedance, is halfway between 90 and 100.

4. We start by calculating the cumulative distribution function, $F(\pi)$, for Euler Fund:

$$\begin{aligned} F(x) &= \int_{-0.10}^{x} f(\pi)d\pi \\ &= c\int_{-0.10}^{x} e^{\pi}d\pi \\ &= c[e^{\pi}]_{-0.10}^{x} \\ &= c(e^{x} - e^{-0.10}) \end{aligned}$$

The value of the CDF at the maximum value of the function must be equal to one. We use this fact to calculate c,

$$F(10\%) = c(e^{0.10} - e^{-0.10}) = 1$$

$$c = \frac{1}{e^{0.10} - e^{-0.10}} = 4.99$$

We then proceed to calculate the one-day 95% VaR, the point where the CDF equals 5%:

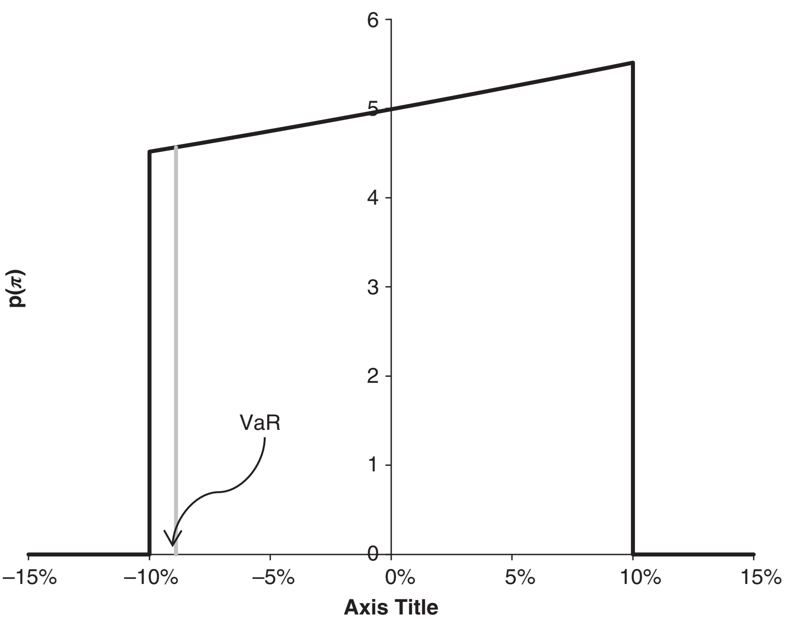
$$F(VaR) = 4.99(e^{-VaR} - e^{-0.10}) = 0.05$$
$$VaR = -\ln\left(\frac{0.05}{4.99} + e^{-0.10}\right)$$
$$VaR = 0.08899$$

VaR is a loss of 8.899%. To find the expected shortfall, S, we proceed as follows:

$$\begin{aligned} S &= -\frac{1}{0.05}\int_{-0.10}^{VaR} \pi f(\pi) d\pi \\ &= -\frac{1}{0.05}\int_{-0.10}^{VaR} \pi c e^{\pi} d\pi \\ &= -\frac{c}{0.05}[e^{\pi}(\pi - 1)]_{-0.10}^{VaR} \\ &= 0.0945 \end{aligned}$$

The expected shortfall for Euler Fund is a loss of 9.45%.

FIGURE EOCQ.1 Euler Fund PDF and VaR



5. (b) Weibull, (c) Fréchet,and (e) Gumbel are distributions associated with extreme value theory.

6. Based on our discussion of translation invariance, we know that cash does not impact the value at risk (VaR) of a portfolio. Based on our discussion of positive homogeneity, we know that \$800 of ABC will have a VaR 8 times the VaR of \$100 of ABC. Therefore, the portfolio with \$200 cash and \$800 of ABC will have a one-day 95% VaR of $8 \times \$4 = \$32$.
7. If we have 1,000 data points in our simulation, then the one-day 99% VaR should correspond to the 10th worst loss, or 11%. The one-day 99% expected shortfall is just the average of the 10 worst losses, or 13%: (17% + 14% + 14% + 13% + 13% + 12% + 12% + 12% + 12% + 11%)/10 = 130%/10 = 13%.
8. If we change only the two worst losses, and the 10th worst loss remains the same, then the one-day 99% VaR is still 11%. We increased each of the two worst losses by 10%, though, so the sum of the 10 worst losses is now 150%. The one-day 99% expected shortfall, the average of the worst losses, is then 150%/10 = 15%.

   Changing the two worst returns had no impact on our estimate of VaR, but had a significant impact on our estimate of the expected shortfall. In practice, there is likely to be considerable uncertainty around extremely rare events. This is a perfect example of why VaR is more robust to outliers compared to expected shortfall.
9. We start by calculating the worst return in each of the eight years, as shown in the last row of the following table.

		Year							
		1	**2**	**3**	**4**	**5**	**6**	**7**	**8**
	1	6%	4%	4%	12%	−1%	−1%	−3%	1%
	2	7%	1%	−3%	2%	−2%	1%	1%	4%
	3	2%	−1%	6%	5%	1%	2%	4%	5%
	4	2%	1%	−5%	−2%	−3%	8%	−4%	3%
	5	−4%	0%	−6%	−2%	−6%	−4%	8%	−10%
	6	1%	−1%	−5%	4%	6%	3%	−15%	6%
Month	**7**	−23%	1%	−2%	−14%	−2%	4%	−13%	−7%
	8	−5%	6%	−1%	−7%	−1%	0%	4%	−9%
	9	1%	0%	8%	−5%	9%	−4%	−6%	9%
	10	6%	1%	5%	−3%	0%	3%	7%	−3%
	11	−1%	−2%	0%	−6%	−4%	−4%	1%	−3%
	12	3%	2%	−3%	0%	4%	0%	5%	−3%
	min	**−23%**	**−2%**	**−6%**	**−14%**	**−6%**	**−4%**	**−15%**	**−10%**

The mean of the minimums is then

$$\text{mean} = \frac{1}{8}(-23\% - 2\% - 6\% - 14\% - 6\% - 4\% - 15\% - 10\%)$$
$$= \frac{-80\%}{8} = -10\%$$

Therefore the expected worst monthly return next year is −10%.

CAPÍTULO 5

1. Using Equation 5.15, we can calculate the portfolio standard deviation,

$$\sigma_p^2 = \$100^2 \times 40\%^2 + (-\$100)^2 \times 30\%^2 + 2 \times \$100 \times (-\$100) \times 37.5\% \times 40\% \times 30\%$$
$$\sigma_p^2 = \$100^2(0.16 + 0.09 - 0.09) = \$100^2(0.16)$$
$$\sigma_p = \$100 \times 40\% = \$40$$

The portfolio standard deviation is \$40. The standard deviation of the AAPL and XOM positions in isolation is \$40 and \$30, respectively. This can be found by multiplying the standard deviation of each security by the absolute exposure of each security.

2. One way to calculate the answer to this problem is to calculate the beta of each security. The dollar beta with respect to the S&P 500 for AAPL is \$100:

$$\beta_{AAPL} = \rho_{AAPL,SPX}\frac{\sigma_{AAPL}}{\sigma_{SPX}} = 50\%\frac{\$40}{20\%} = \$100$$

If the correlation between XOM and the S&P 500 is 80%, then the correlation between a short position in XOM and the S&P 500 is going to be −80%. The beta of XOM is then −\$120:

$$\beta_{XOM} = \rho_{XOM,SPX}\frac{\sigma_{XOM}}{\sigma_{SPX}} = -80\%\frac{\$30}{20\%} = -\$120$$

We can find the beta of the entire portfolio by adding the two position betas together. The beta of the entire portfolio is \$100 + (−\$120) = −\$20. To minimize the variance of the portfolio we would then buy \$20 of the S&P 500.

3. We can find the answer by substituting the values provided into the regression equation.

$$\begin{aligned} \mathrm{E}[r_{ABC}|r_A, r_B] &= 0.01 + 1.25 \times r_A + 0.34 \times r_B \\ &= 0.01 + 1.25 \times 0.10 + 0.34 \times 0.50 \\ &= 0.01 + 0.125 + 0.17 \\ &= 0.305 \end{aligned}$$

The final answer is 30.5%.

4. Covariance = 4.87%; correlation = 82.40%.
5. First we note that the expected value of X_A plus X_B is just the sum of the means:

$$E[X_A + X_B] = E[X_A] + E[X_B] = \mu_A + \mu_B$$

Substituting into our equation for variance, and rearranging, we get

$$\begin{aligned}\sigma^2_{A+B} &= E[(X_A + X_B - E[X_A + X_B])^2] \\ &= E[((X_A - \mu_A) + (X_B - \mu_B))^2]\end{aligned}$$

Expanding the squared term and solving,

$$\begin{aligned}\sigma^2_{A+B} &= E[(X_A - \mu_A)^2 + (X_B - \mu_B)^2 + 2(X_A - \mu_A)(X_B - \mu_B)] \\ &= E[(X_A - \mu_A)^2] + E[(X_B - \mu_B)^2 + 2E[(X_A - \mu_A)(X_B - \mu_B)]]\end{aligned}$$

The first and second terms on the right-hand side are the variance of A and B, respectively. The last term is the covariance between them. Therefore,

$$\sigma^2_{A+B} = \sigma^2_A + \sigma^2_B + 2\mathrm{Cov}[X_A, X_B]$$

Using our definition of covariance, we can rewrite the covariance in terms of correlation and standard deviation, arriving at our final answer,

$$\sigma^2_{A+B} = \sigma^2_A + \sigma^2_B + 2\rho_{AB}\sigma_A\sigma_B$$

6. From our definition of R^2, we have

$$R^2 = 1 - \frac{\mathrm{RSS}}{\mathrm{TSS}} = 1 - \frac{10.80\%}{13.50\%} = 20\%$$

The R^2 is 20%.

7. Using Equation 5.50, the corresponding F-statistic is,

$$F = \frac{R^2/(n-1)}{(1-R^2)/(t-n)} = \frac{20\%/(2-1)}{(1-20\%)/(50-2)} = 12$$

Using a spreadsheet or other program, we see that the probability associated with this F-statistic is 0.11%; that is, there is only a 0.11% chance that an F-statistic of this magnitude (or greater) would have happened by chance. The F-statistic is significant at the 95% confidence level.

8. We want to start by rewriting the equation for the covariance of X and Y,

$$\mathrm{Cov}[X, Y] = E[(X - E[X])(Y - E[Y])]$$

Using our linear regression equation and making use of the OLS assumptions, we see that the second term can be expressed in terms of X, β, and ε:

$$\begin{aligned} Y - \mathrm{E}[Y] &= (\alpha + \beta X + \varepsilon) - \mathrm{E}[\alpha + \beta X + \varepsilon] \\ &= (\alpha + \beta X + \varepsilon) - (\alpha + \beta \mathrm{E}[X] + \mathrm{E}[\varepsilon]) \\ &= \beta(X - \mathrm{E}[X]) + (\varepsilon - \mathrm{E}[\varepsilon]) \end{aligned}$$

In the last line, because it is equal to zero, we could have removed the term $\mathrm{E}[\varepsilon]$, but, as we will see, keeping this last term as it is, which expresses ε as a deviation from the mean, will be useful.

Substituting into our covariance equation,

$$\begin{aligned} \mathrm{Cov}[X, Y] &= \mathrm{E}[(X - \mathrm{E}[X])(\beta(X - \mathrm{E}[X]) + (\varepsilon - \mathrm{E}[\varepsilon]))] \\ &= \mathrm{E}[\beta(X - \mathrm{E}[X])^2 + (X - \mathrm{E}[X])(\varepsilon - \mathrm{E}[\varepsilon])] \\ &= \beta \mathrm{E}[(X - \mathrm{E}[X])^2] + \mathrm{E}[(X - \mathrm{E}[X])(\varepsilon - \mathrm{E}[\varepsilon])] \end{aligned}$$

The first expectation on the right-hand side is the variance of X, and the last term is the covariance of X and ε. We then have

$$\mathrm{Cov}[X, Y] = \beta\sigma_X^2 + \mathrm{Cov}[X, \varepsilon]$$

As we stated with Equation 5.28, one of the assumption of the OLS model is that X and ε are uncorrelated; therefore,

$$\mathrm{Cov}[X, Y] = \beta\sigma_X^2$$

All that remains is to divide both sides by the variance of X, and to expand the correlation term,

$$\beta = \frac{\mathrm{Cov}[X, Y]}{\sigma_X^2} = \frac{\rho_{XY}\sigma_X\sigma_Y}{\sigma_X^2} = \rho_{XY}\frac{\sigma_Y}{\sigma_X}$$

9. We can use our Cholesky decomposition algorithm to find the elements of the matrix

$$\begin{aligned} l_{11} &= \sqrt{\sigma_{11}} = \sqrt{4} = 2 \\ l_{21} &= \frac{1}{l_{11}}\sigma_{21} = \frac{1}{2}14 = 7 \\ l_{22} &= \sqrt{\sigma_{22} - l_{21}^2} = \sqrt{50 - 7^2} = 1 \\ l_{31} &= \frac{1}{l_{11}}\sigma_{31} = \frac{1}{2}16 = 8 \\ l_{32} &= \frac{1}{l_{22}}\left(\sigma_{32} - l_{31}l_{21}\right) = \frac{1}{1}(58 - 8 \times 7) = 2 \\ l_{33} &= \sqrt{\sigma_{33} - l_{31}^2 - l_{32}^2} = \sqrt{132 - 8^2 - 2^2} = 8 \end{aligned}$$

We can express the full lower triangular matrix as:

$$\mathbf{L} = \begin{bmatrix} 2 & 0 & 0 \\ 7 & 1 & 0 \\ 8 & 2 & 8 \end{bmatrix}$$

We can verify this answer by noting that $\mathbf{LL}'$ is indeed equal to our original covariance matrix, Σ.

10. The expected return of XYZ is 6.01%,

$$\begin{aligned} \mathrm{E}[r_{\mathrm{XYZ}}|r_{\mathrm{index}}] &= \mathrm{E}[\alpha + \beta r_{\mathrm{index}} + \varepsilon|r_{\mathrm{index}}] \\ &= \alpha + \beta r_{\mathrm{index}} \\ &= 0.01\% + 1.20 \times 5.0\% \\ &= 6.01\% \end{aligned}$$

11. The expected value of r_{XYZ} is 0.07%,

$$\begin{aligned} \mathrm{E}[r_{\mathrm{XYZ}}] &= \mathrm{E}[\alpha + \beta r_{\mathrm{index}} + \varepsilon] \\ &= \alpha + \beta E[r_{\mathrm{inde}}] \\ &= 0.01\% + 1.20 \times 0.05\% \\ &= 0.07\% \end{aligned}$$

To calculate the variance of r_{XYZ}, we start by re-expressing the deviation from the mean for r_{XYZ}. If

$$r_{\mathrm{XYZ}} = \alpha + \beta r_{\mathrm{index}} + \varepsilon$$

and

$$\mathrm{E}[r_{\mathrm{XYZ}}] = \alpha + \beta \mathrm{E}[r_{\mathrm{index}}] + \mathrm{E}[\varepsilon]$$

then

$$r_{\mathrm{XYZ}} - \mathrm{E}[r_{\mathrm{XYZ}}] = \beta(r_{\mathrm{index}} - \mathrm{E}[r_{\mathrm{index}}]) + (\varepsilon - \mathrm{E}[\varepsilon])$$

The variance of r_{XYZ} is, then

$$\begin{aligned} \sigma^2_{\mathrm{XYZ}} &= \mathrm{E}[(r_{\mathrm{XYZ}} - \mathrm{E}[r_{\mathrm{XYZ}}])^2] \\ &= \mathrm{E}[(\beta(r_{\mathrm{index}} - \mathrm{E}[r_{\mathrm{index}}]) + (\varepsilon - \mathrm{E}[\varepsilon]))^2] \\ &= \mathrm{E}[\beta^2(r_{\mathrm{index}} - \mathrm{E}[r_{\mathrm{index}}]) + 2\beta(\varepsilon - \mathrm{E}[\varepsilon])(r_{\mathrm{index}} - \mathrm{E}[r_{\mathrm{index}}]) + (\varepsilon - \mathrm{E}[\varepsilon])^2] \\ &= \beta^2 \mathrm{E}[(r_{\mathrm{index}} - \mathrm{E}[r_{\mathrm{index}}])] + 2\beta \mathrm{E}[(\varepsilon - \mathrm{E}[\varepsilon])(r_{\mathrm{index}} - \mathrm{E}[r_{\mathrm{index}}])] \\ &\quad + \mathrm{E}[(\varepsilon - \mathrm{E}[\varepsilon])^2] \end{aligned}$$

$$
\begin{aligned}
&= \beta\sigma_{\text{index}}^2 + 2\beta \mathrm{Cov}[r_{\text{index}}, \varepsilon] + \sigma_\varepsilon^2 \\
&= \beta\sigma_{\text{index}}^2 + \sigma_\varepsilon^2 \\
&= 0.000424
\end{aligned}
$$

To get to the second to last line, we use the fact that the covariance between the regressor and the disturbance term is zero in a linear regression.

Taking the square root of the variance, we get a standard deviation of 2.06%.

12. We start by re-expressing the deviation from the mean for r_{XYZ},similar to the previous question:

$$r_{\text{XYZ}} - \mathrm{E}[r_{\text{XYZ}}] = \beta(r_{\text{index}} - \mathrm{E}[r_{\text{index}}]) + (\varepsilon - \mathrm{E}[\varepsilon])$$

We then substitute this into the equation for covariance,

$$
\begin{aligned}
\mathrm{Cov}[r_{\text{XYZ}}, r_{\text{index}}] &= \mathrm{E}[(r_{\text{XYZ}} - \mathrm{E}[r_{\text{XYZ}}])(r_{\text{index}} - \mathrm{E}[r_{\text{index}}])] \\
&= \mathrm{E}[(\beta(r_{\text{index}} - \mathrm{E}[r_{\text{index}}]) + (\varepsilon - \mathrm{E}[\varepsilon]))(r_{\text{index}} - \mathrm{E}[r_{\text{index}}])] \\
&= \mathrm{E}[\beta(r_{\text{index}} - \mathrm{E}[r_{\text{index}}])^2 + (r_{\text{index}} - \mathrm{E}[r_{\text{index}}])(\varepsilon - \mathrm{E}[\varepsilon])] \\
&= \beta \mathrm{E}[(r_{\text{index}} - \mathrm{E}[r_{\text{index}}])^2] + \mathrm{E}[(r_{\text{index}} - \mathrm{E}[r_{\text{index}}])(\varepsilon - \mathrm{E}[\varepsilon])]
\end{aligned}
$$

The first term on the right-hand side is the variance of r_{index}, while the second term is the covariance of r_{index} and the ε, which is zero. This gives us

$$\mathrm{Cov}[r_{\text{XYZ}}, r_{\text{index}}] = \beta\sigma_{\text{index}}^2$$

The correlation is then

$$\rho = \frac{\mathrm{Cov}[r_{\text{XYZ}}, r_{\text{index}}]}{\sigma_{\text{XYZ}}\sigma_{\text{index}}} = \frac{\beta\sigma_{\text{index}}^2}{\sigma_{\text{XYZ}}\sigma_{\text{index}}} = \beta\frac{\sigma_{\text{index}}}{\sigma_{\text{XYZ}}} = 1.20\frac{1.50\%}{2.06\%} = 87.42\%$$

CAPÍTULO 6

1. If you examine the data closely, you'll notice that Fund A and Fund B have both had the same returns, only in different orders. Because of this, the mean and standard deviation of their returns must be equal. We start by finding those quantities. Remember, we were asked to find the population coskewness, so we should calculate the populate standard deviation not the sample standard deviation. Also, because multiplying one or both variables by a constant will not change the coskewness, rather than using the percent values we use the values multiplied by 100 (so, 3 instead of 0.03). We have

$$\mu_A = \mu_B = \frac{1}{5}[-3 - 1 + 0 + 1 + 3] = 0$$

$$\sigma_A^2 = \sigma_B^2 = \frac{1}{5}[(-3-0)^2 + (-1-0)^2 + (0-0)^2 + (1-0)^2 + (3-0)^2]$$
$$= \frac{1}{5}[9+1+0+1+9]$$
$$= 4$$
$$\sigma_A = \sigma_B = 2$$

Next, we need to calculate the cross-central moment,

$$\mu_{AAB} = \frac{1}{5}[(-3-0)^2(-3-0) + (-1-0)^2(0-0)$$
$$+ (0-0)^2(3-0) + (1-0)^2(1-0) + (3-0)^2(-1-0)]$$
$$= \frac{1}{5}[-27+0+0+1-9]$$
$$= -7$$

Putting it all together, we arrive at our final answer, −7/8,

$$S_{AAB} = \frac{\mu_{AAB}}{\sigma_A^2\sigma_B} = -\frac{7}{4\times 2} = -\frac{7}{8}$$

2. Because the bonds are independent, and only because the bonds are independent, the probability of both bonds defaulting is the probability of one bond defaulting multiplied by the probability of the other bond defaulting. If we represent the first bond defaulting by A and the second by B, then in this case P[A and B] = P[A] × P[B] = 10% × 10% = 1%. Remember, we can only do this because A and B are independent. Similarly, the probability of neither bond defaulting is, P[$\overline{A}$ and $\overline{B}$] = (1 − P[A])(1 − P[B]) = 90% × 90% = 81%, where $\overline{A}$ can be read "not A." The probability of one bond defaulting is 18%. There are two ways for only one bond to default: A can default and B does not, or B can default and A does not. The probability of each of the events is 9%: P[$\overline{A}$ and B] = (1 − P[A])P[B] = 90% × 10% = 9%, and P[A and $\overline{B}$] = P[A] (1 − P[B]) = 10% × 90% = 9%. Therefore, the probability of just one bond defaulting is 9% + 9% = 18%. This is easy to see in a probability matrix.

		Bond A	
		Default	**No Default**
Bond B	**Default**	1%	9%
	No Default	9%	81%

As is required, the sum of all possible outcomes is equal to 100%, 1% + 81% + 18% = 100%.

3. We can calculate the probability that Option A ends up in the money by adding the up the probabilities in the left column of the probability matrix, $P[A_{ITM}] = 40\% + 20\% = 60\%$. Similarly, the probability that Option B is in the money is found by adding the entries in the first row $P[B_{ITM}] = 40\% + 10\% = 50\%$.

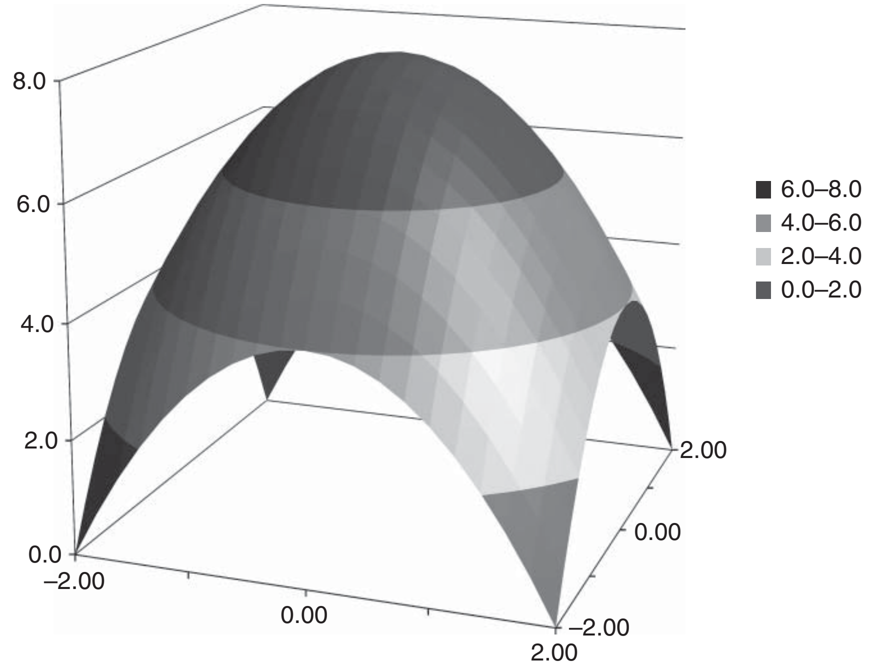
The probability that both options end up in the money is $P[B_{ITM} \text{ and } A_{ITM}] = 40\%$. If the options were independent, we would expect this joint probability to be $60\% \times 50\% = 30\%$. Therefore, Option A and B are not independent, $P[B_{ITM} \text{ and } A_{ITM}] \neq P[B_{ITM}] \times P[A_{ITM}]$.

4. The shape of the PDF resembles a truncated paraboloid, as shown in Figure 6.A1.

To see if X and Y are independent, we start by calculating the marginal distribution of X,

$$\begin{aligned} f_x(x) &= \int_y f(x, t)dt \\ &= \int_{-2}^{+2} c(8 - x^2 - t^2)dt \\ &= c\left[8t - tx^2 - \frac{1}{3}t^3\right]_{-2}^{+2} \\ &= 4c\left(\frac{20}{3} - x^2\right) \end{aligned}$$

FIGURE 6.A1 Joint Probability Density Function



The marginal distribution of Y is

$$\begin{aligned} f_y(y) &= \int_x f(t,y)dt \\ &= \int_{-2}^{+2} c(8 - t^2 - y^2)dt \\ &= c\left[8t - ty^2 - \frac{1}{3}t^3\right]_{-2}^{+2} \\ &= 4c\left(\frac{20}{3} - y^2\right) \end{aligned}$$

Putting the two together, we can see that the product of the marginal distributions does not equal the joint distribution.

$$\begin{aligned} f_x(x)f_y(y) &= 16c^2\left(\frac{20}{3} - x^2\right)\left(\frac{20}{3} - y^2\right) \\ &= \frac{1}{16^3}(400 - 60x^2 - 60y^2 + 9x^2y^2) \\ &\neq f(x,y) \end{aligned}$$

We conclude that X and Y are *not* independent.

5. The joint distribution in 6.10a is nonelliptical and the two variables are negatively correlated. The joint distribution in 6.10b is elliptical and the two variables are positively correlated.

6. To calculate Kendall's tau, we start by examining the concordance of all possible pairs of points. With three points there are three distinct pairs:

Pair	**Concordant = +1, Discordant = −1**
A, B	−1
A, C	−1
B, C	+1

One pair is concordant and two are discordant; therefore, Kendall's tau is −33%,

$$\tau = \frac{\#\text{ of concordant points} - \#\text{ of discordant points}}{\binom{n}{2}} = \frac{1-2}{3} = -33\%$$

To calculate Spearman's rho, we start by calculating the rank of each data point,

	X	Y	Rank[X]	Rank[Y]
A	70%	5%	1	3
B	40%	35%	2	1
C	20%	10%	3	2

The general formula for correlation is

$$\text{Corr}[X, Y] = \frac{\sum_{i=1}^{n} (x_i - \bar{x})\left(y_i - \bar{y}\right)}{\sqrt{\sum_{i=1}^{n} (x_i - \bar{x})^2}\sqrt{\sum_{i=1}^{n} \left(y_i - \bar{y}\right)^2}}$$

where $\bar{x}$ and $\bar{y}$ are the mean of X and Y respectively. To calculate Spearman's rho, we use not X and Y values, but their ranks. The mean of both Rank[X] and Rank[Y] is 2; therefore,

$$\begin{aligned}
\rho_s &= \frac{(1-2)(3-2)+(2-2)(1-2)+(3-2)(2-2)}{\sqrt{(1-2)^2+(2-2)^2+(3-2)^2}\sqrt{(3-2)^2+(1-2)^2+(2-2)^2}} \\
&= \frac{(-1)+0+0}{\sqrt{1+0+1}\sqrt{1+1+0}} \\
&= \frac{-1}{\sqrt{2}\sqrt{2}} \\
&= -50\%
\end{aligned}$$

Our final answers are then that Kendall's tau is −33% and Spearman's rho is −50%. In this example, Spearman's rho and Kendall's tau share the same sign, but they are not equal.

7. We start by calculating the copula's density function, $c(u, v)$:

$$c(u,v) = \frac{\partial^2 C(u,v)}{\partial u \partial v} = \frac{\partial^2}{\partial u \partial v} uv = \frac{\partial}{\partial u} u = 1$$

We next calculate the expected value of the copula:

$$\begin{aligned}
\int_0^1 \int_0^1 C(u,v)c(u,v)dudv &= \int_0^1 \int_0^1 uvdudv \\
&= \int_0^1 \left[\frac{1}{2}u^2 v\right]_0^1 dv \\
&= \frac{1}{2}\int_0^1 vdv = \frac{1}{2}\left[\frac{1}{2}v^2\right]_0^1 = \frac{1}{4}
\end{aligned}$$

Substituting into our equation for Kendall's tau,

$$\tau = 4\int_0^1\int_0^1 C(u,v)c(u,v)dudv - 1 = 4\frac{1}{4} - 1 = 0$$

Not surprisingly for something called the independent copula, Kendall's tau is zero.

8. To calculate Spearman's rho, we first integrate the copula, $C(u,v)$, function with respect to u:

$$\begin{aligned}
\int_0^1 C(u,v)du &= \int_0^1 uv[1+\alpha(1-u)(1-v)]du \\
&= \int_0^1 [u(v+\alpha v(1-v)) - u^2\alpha v(1-v)]du \\
&= \left[\frac{1}{2}u^2(v+\alpha v(1-v)) - \frac{1}{3}u^3\alpha v(1-v)\right]_0^1 \\
&= \left[\frac{1}{2}(v+\alpha v(1-v)) - \frac{1}{3}\alpha v(1-v)\right] - [0-0] \\
&= \left(\frac{1}{2}+\frac{1}{6}\alpha\right)v - \frac{1}{6}\alpha v^2
\end{aligned}$$

Next, we integrate this result with respect to v:

$$\begin{aligned}
\int_0^1\int_0^1 C(u,v)dudv &= \int_0^1 \left[\left(\frac{1}{2}+\frac{1}{6}\alpha\right)v - \frac{1}{6}\alpha v^2\right]dv \\
&= \left[\frac{1}{2}\left(\frac{1}{2}+\frac{1}{6}\alpha\right)v^2 - \frac{1}{18}\alpha v^3\right]_0^1 \\
&= \left[\frac{1}{2}\left(\frac{1}{2}+\frac{1}{6}\alpha\right) - \frac{1}{18}\alpha\right] - [0-0] \\
&= \frac{1}{4} + \frac{1}{36}\alpha
\end{aligned}$$

Finally, we use this result to calculate Spearman's rho:

$$\begin{aligned}
\rho &= 12\int_0^1\int_0^1 C(u,v)dudv - 3 \\
&= 12\left(\frac{1}{4}+\frac{1}{36}\alpha\right) - 3 \\
&= \frac{1}{3}\alpha
\end{aligned}$$

9. The daily log returns are independent and identically distributed. Because these are log returns, each annual log return would just be the sum of the 256 daily log returns. For the annual log returns, then, the mean, standard deviation, skewness, and excess kurtosis would be 25.6%, 16%, −0.10, and 0.02, respectively.

$$\mu_A = 256 \times 0.10\% = 25.6\%$$

$$\sigma_A = \sqrt{256} \times 1\% = 16\%$$

$$S_A = -\frac{1.6}{\sqrt{256}} = -0.10$$

$$K_{\mathrm{Ex},A} = \frac{5.12}{256} = 0.02$$

CAPÍTULO 7

1. The VaR of the portfolio would be approximately \$10.6 million. Using Equation 7.5:

$$d(\mathrm{VaR}) = \frac{dw_i}{w_i}\mathrm{iVaR}_i = \frac{\$1.2\ million - \$1.0\ million}{\$1.0\ million}\$3\ million = 0.2 \times \$3\ million$$

$$d(\mathrm{VaR}) = \$0.6\ million$$

The total VaR would then be approximately \$10.6 million, \$10 million + \$0.6 million = \$10.6 million.

2. The iVaR of Position B is −\$100. For the portfolio as a whole, the iVaR is equal to the VaR, so the iVaR of this portfolio must be \$1,000. For any portfolio the sum of the iVaRs must be equal to the total VaR or iVaR. In this case \$300 + (−\$100) + \$800 = \$1,000. Remember, unlike VaR, iVaR can be negative. The fact that the iVaR of Position B is negative tells us that increasing the size of Position B, at least at the margin, will actually lower the overall risk of the portfolio.

	VaR	**iVaR**
Position A	\$500	\$300
Position B	\$400	−\$100
Position C	\$650	\$800
Portfolio	\$1,000	\$1,000

3. In order to calculate the variance of the portfolio we first calculate the expected value of r_p:

$$\begin{aligned} \mathrm{E}[r_p] &= \mathrm{E}[\alpha] + E[\beta r_{Index}] + \mathrm{E}[\varepsilon] \\ &= \alpha + \beta E[r_{Index}] + \mathrm{E}[\varepsilon] \end{aligned}$$

The portfolio variance, σ_p^2, is then

$$\begin{aligned} \sigma_p^2 &= \mathrm{E}[(r_p - \mathrm{E}[r_p])^2] = \mathrm{E}[(\alpha + \beta r_{Index} + \varepsilon - \alpha - \beta \mathrm{E}[r_{Index}] - \mathrm{E}[\varepsilon])^2] \\ &= \mathrm{E}[(\beta(r_{Index} - \mathrm{E}[r_{Index}]) + \varepsilon - \mathrm{E}[\varepsilon])^2] \end{aligned}$$

$$= \mathrm{E}[\beta^2(r_{Index} - E[r_{Index}])^2 + 2(r_{Index} - E[r_{Index}])(\varepsilon - \mathrm{E}[\varepsilon]) + (\varepsilon - \mathrm{E}[\varepsilon])^2]$$
$$= \beta^2 E[(r_{Index} - E[r_{Index}])^2] + 2E[(r_{Index} - E[r_{Index}])(\varepsilon - \mathrm{E}[\varepsilon])] + E[(\varepsilon - \mathrm{E}[\varepsilon])^2]$$
$$= \beta^2 \sigma^2_{Index} + 2 \times 0 + \sigma^2_\varepsilon$$
$$= \beta^2 \sigma^2_{Index} + \sigma^2_\varepsilon$$

In the last line we have expressed the total portfolio variance as a linear combination of the variance of the index, σ^2_{Index}, and the variance of the error term, σ^2_ε. As anticipated, for variance the sum of systematic and idiosyncratic risk equals the total portfolio risk.

For standard deviation, the total portfolio risk is generally less than the sum of the systematic and idiosyncratic risk. They are only equal in the degenerate cases when the systematic risk, the idiosyncratic risk, or both are zero. Mathematically,

$$\sigma_p \leq \beta\sigma_{Index} + \sigma_\varepsilon$$

We can prove this by completing the sum of squares under the square root. Standard deviation must be positive; therefore, if β is positive, then $\sigma_{Index}\sigma_\varepsilon > 0$, and

$$\sqrt{\beta^2\sigma^2_{Index} + \sigma^2_\varepsilon} \leq \sqrt{\beta^2\sigma^2_{Index} + 2\beta\sigma_{Index}\sigma_\varepsilon + \sigma^2_\varepsilon}$$

or,

$$\sqrt{\beta^2\sigma^2_{Index} + \sigma^2_\varepsilon} \leq \beta\sigma_{Index} + \sigma_\varepsilon$$

From our previous result, the left-hand side is equal to the portfolio standard deviation, so

$$\sigma_p \leq \beta\sigma_{Index} + \sigma_\varepsilon$$

4. $h = 0.50$ or 50%.

$$h = 1 - \frac{30}{10 + 20 + 30} = 1 - \frac{30}{60} = 0.50$$

5. $N = 2.45$

$$N = e^{-[0.60 \times \ln(0.60) + 0.30 \times \ln(0.30) + 0.10 \times \ln(0.10)]} = e^{0.90} = 2.45$$

6. If you can borrow at the risk-free rate, then applying leverage will not impact the Sharpe ratio. The Sharpe ratio of the 3× version of the ETF will also have a Sharpe ratio of 0.80.

The expected return of the levered product is 50%. Rearranging Equation 7.14, we have

$$R_i = S_i\sigma_i + R_{\mathrm{rf}}$$

For the ETF,

$$R_{\mathrm{ETF}} = 0.80 \times 20\% + 2\% = 18\%$$

To create the levered product, for each \$1 invested, we need to borrow an additional \$2, so that we can buy \$3 of the ETF. The return would then be three times the return of the ETF minus two times the risk-free rate:

$$R_{3\times} = 3 \times 18\% - 2 \times 2\% = 50\%$$

We can double check this using Equation 7.14. The standard deviation of the 3× product would be three times the standard deviation of the ETF, or 60%, therefore,

$$S_{3\times} = \frac{50\% - 2\%}{60\%} = 0.80$$

As required, the Sharpe ratio of the levered product is the same as for the ETF.

In practice, any return, positive or negative, will change the leverage ratio of a levered product. In order to maintain a constant level of leverage, a levered product would need to be rebalanced from time to time. The standard deviation of the underlying, how often the product is rebalanced, and the cost of rebalancing will all impact the actual performance of a levered product.

7. Using Equation 7.15, we calculate an incremental Sharpe of 0.50 for Alice and 0.20 for Bob,

$$S_A^* = 0.80 - 0.30 \times 1.00 = 0.50$$

$$S_B^* = 1.10 - 0.90 \times 1.00 = 0.20$$

Both managers would improve the Sharpe ratio of the fund at the margin, but Alice would improve the Sharpe ratio more.

CAPÍTULO 8

1. The present value is \$1,092.97. Using Equation 8.1, we have

$$V_0 = \frac{10\% \times \$1{,}000}{(1+5\%)} + \frac{10\% \times \$1{,}000}{(1+5\%)^2} + \frac{\$1{,}000}{(1+5\%)^2}$$
$$= \$95.24 + \$90.70 + \$907.03 = \$1{,}092.97$$

2. If there is no default, the bondholder will receive \$104 at the end of the year, the \$4 coupon plus the \$100 notional. If the bond defaults, the bondholder will receive 20% of this amount. The risk-neutral value of the bond is just the weighted average of these two values discounted back at the risk-free rate:

$$V_0 = 88\% \frac{\$104}{1.04} + 12\% \frac{20\% \times \$104}{1.04}$$
$$= \frac{\$104}{1.04}(88\% + 12\% \times 20\%)$$

$$= \$100(88\% + 2.4\%)$$
$$= \$100 \times 90.4\%$$
$$= \$90.40$$

The risk-neutral value is $90.40.

3. This setup is simple enough that we can find an explicit solution for the yield. The yield, Y, must satisfy the following equation:

$$V_0 = \$85 = \frac{\$10}{(1+Y)} + \frac{\$110}{(1+Y)^2}$$

Rearranging, we have

$$\$85(1+Y)^2 - \$10(1+Y) - \$110 = 0$$

This is a quadratic equation. The solution can then be found as

$$\begin{aligned}(1+Y) &= \frac{\$10 \pm \sqrt{\$10^2 + 4 \times \$85 \times \$110}}{2 \times \$85} \\ &= \frac{\$10 \pm \$193.65}{2 \times \$85} = 5.88\% \pm 113.91\% \\ &= 119.79\%\end{aligned}$$

In the last line, we selected the positive root, which gives us a yield of 19.79%. The negative root, while a solution to the quadratic equation, does not make much sense for the yield. The yield is 19.79%.

4. The discount rate used by investors, R, must satisfy the following equation:

$$\$72 = (1 - 19\%)\frac{\$100}{1+R} + 19\%\frac{(1 - 100\%)\$100}{1+R}$$

Rearranging, and solving for R:

$$\$72 = (1 - 19\%)\frac{\$100}{1+R} = \frac{\$81}{1+R}$$
$$R = \frac{\$81}{\$72} - 1 = \frac{9}{8} - 1 = \frac{1}{8} = 12.5\%$$

Compared to the risk-free rate of 5%, the investors are asking for an additional 7.5% to accept the credit risk of this bond. The correct answer is 7.5%.

5. The enterprise value is equal to $4,000 = $1,924 + $2,076. (The value of the equity is equal to the market capitalization.) Using Equation 8.15,

$$\Delta = \frac{-\ln\left(\frac{V_E}{B}\right) - \left(r - \frac{\sigma_V^2}{2}\right)T}{\sigma_V\sqrt{T}} = \frac{-\ln\left(\frac{\$4{,}000}{\$2{,}076}\right) - 0}{40\%\sqrt{1}} = -1.64$$

The distance to default is −1.64 standard deviations. Plugging this into the standard normal cumulative distribution, we get our final answer for the probability of default, 5%.

6. There is a 9.01% probability that exactly four bonds default:

$$P[K = k] = \binom{n}{k} p^k (1 - p)^{n-k}$$

$$P[K = k] = \binom{40}{4} 5\%^4 95\%^{36} = 9.01\%$$

CAPÍTULO 9

1. In terms of shares, the 15-day average trading volume based on the mean is 28.9 million shares. Based on the median, the average is 22 million shares.

   To calculate the average dollar volume, we first need to calculate the dollar volume each day. We do this by multiplying the price and volume figures each day, as in the following table. Based on this new column, the 15-day mean volume is \$4.4 billion. Based on the median, it is \$3.4 billion.

   In both cases, the mean and median are considerably different. You can get a sense of how much the volume varies by comparing the volume on June 8th and 12th.

	Volume (millions of shares)	**Price ($)**	**Volume ($millions)**
5/23/2017	20	153.80	3,076.00
5/24/2017	19	153.34	2,913.46
5/25/2017	19	153.87	2,923.53
5/26/2017	22	153.61	3,379.42
5/30/2017	20	153.67	3,073.40
5/31/2017	24	152.76	3,666.24
6/1/2017	16	153.18	2,450.88
6/2/2017	28	155.45	4,352.60
6/5/2017	25	153.93	3,848.25
6/6/2017	27	154.45	4,170.15
6/7/2017	21	155.37	3,262.77
6/8/2017	21	154.99	3,254.79
6/9/2017	65	148.98	9,683.70
6/12/2017	72	145.42	10,470.24
6/13/2017	34	146.59	4,984.06

2. You own 0.5 days' volume of IBM:

$$2 \text{ milllion shares} \frac{\text{day}}{4 \text{ million shares}} = 0.5 \text{ days}$$

3. Because we are assuming 20% of average daily volume, we can sell 0.8 million shares of IBM and cover1.2 million shares of MSFT each day. At current market prices this is equal to \$64 million per day of IBM and \$60 million per day of MSFT. Our current portfolio contains a total of \$160 million of IBM and −\$150 million of MSFT. The complete liquidity schedule is as follows:

	Per Day			Cumulative	
Day	IBM (\$MM)	MSFT (\$MM)	Total (\$MM)	Total (\$MM)	% Liquidated
1	\$64	\$60	\$124	\$124	40%
2	\$64	\$60	\$124	\$248	80%
3	\$32	\$30	\$62	\$310	100%

4. If there are 256 business days in the year, and the annualized volatility is 16%, then the daily volatility is $1\% = 16\%/\sqrt{256}$. If returns are normally distributed, then we need to multiply this daily standard deviation by 1.645. Using Equation 9.2, we have

$$\text{LVaR}_{\text{ABC}} = 1{,}000 \times 1 \times 1\% \times 1.645 + 1{,}000 \times 0.01 = 26.45$$

$$\text{LVaR}_{\text{XYZ}} = 1 \times 1{,}000 \times 1\% \times 1.645 + 1 \times 1 = 17.45$$

The LVaR of \$1,000 invested in ABC or XYZ are \$26.45 and \$17.45, respectively. You could also have expressed your answer as a percentage of total market value, in which case the answer would be 2.64% and 1.74%, respectively.

The VaR of both positions is the same, but the LVaR of the ABC is greater. This is because the bid-ask spread, even though it is less in dollar terms, is greater in percentage terms.

5. We can rearrange Equation 9.3 to get the approximate change in price:

$$dP = \frac{1}{\lambda}\frac{dQ}{Q}P = \frac{1}{-4}\frac{2.5 \text{ million}}{10 \text{ million}}\$64 = -\frac{1}{4}\frac{1}{4}\$64 = -\$4$$

Trying to sell an additional 2.5 million shares will cause the price to fall by approximately \$4, giving us a final price for XYZ of \$60. For the sale, you can assume you will get a price close to the mean of the starting price and the final price, or \$62. The total proceeds would then be \$155 million.

In reality, Equation 9.3 is only true for infinitesimal changes. In this case, though, the approximation is very close. We could have used Equation 9.4 to calculate the exact price change. If we had done this, rather than \$60, the exact final price would have been \$60.53.